1. Hindcast Functional Plant Traits

with multispectral imagery and recent imaging spectroscopy

Multispectral imagery (Landsat, Sentinel)

Imaging spectroscopy (BioSCape, SHIFT, AVIRIS)

Field data (plots, traits)

Annual maps of community mean traits (1984 – present – 2050)

1984 ... **Present** ... **2050 (forecast)**

Community mean traits (per pixel, per year)

Leaf N · LMA · Water content · Pigments · ...

2. Infer Species Mixtures & Trait Distributions

Observed trait vector
(at each pixel)

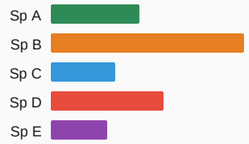
- Leaf N
- LMA
- Water content
- Pigments

Species trait matrix
(species × traits)

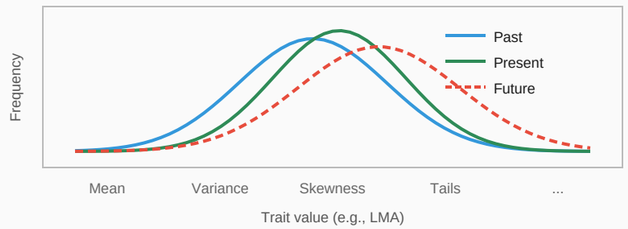
	Leaf N	LMA	Water	Pigments
Sp A	2.1	120	0.62	0.41
Sp B	1.4	180	0.48	0.55
Sp C	2.8	95	0.71	0.33
Sp D	1.9	150	0.55	0.60
Sp E	3.2	110	0.66	0.28

Maximum Entropy
Least biased feasible community

Inferred species mixture (relative abundances at each pixel)



Community trait distributions through time (at each pixel)



3. Trait Driver Theory & Ecological Change

Test mechanisms and forecast ecosystem reorganization

Core processes (Trait Driver Theory)

Selection
(environmental filtering)

Shifts in trait optima and niche breadth

Dispersal
(movement, colonization)

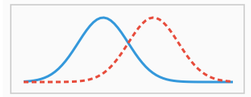
Introduces new traits and modes

Drift
(stochastic dynamics)

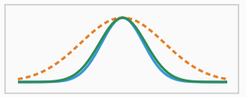
Random changes in trait frequencies

Test key hypotheses

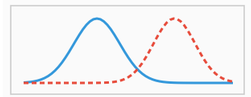
Environmental filtering
Trait optima shift under climate change



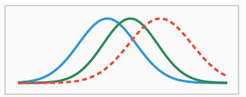
Post-fire succession
Distributions broaden then narrow



Range expansion
New modes as species arrive



Biome reorganization
Community-level shifts in trait distributions



Ecological change (forecast)

- Community turnover
- Functional diversity
- Resilience
- Regime shifts
- Novel communities

Remote sensing observations → **Reconstructed community trait distributions** → **Mechanistic understanding and forecasting**